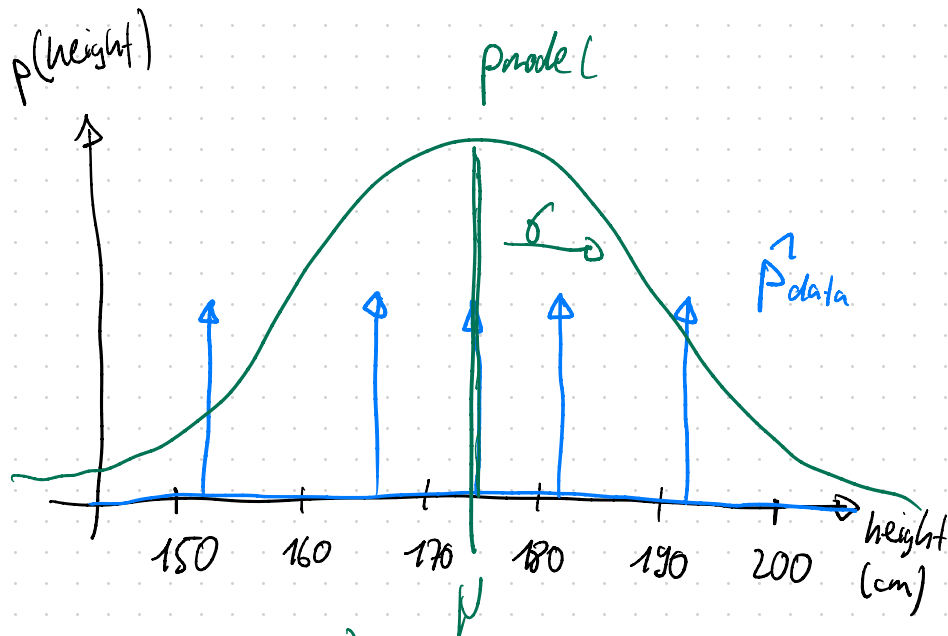


Example

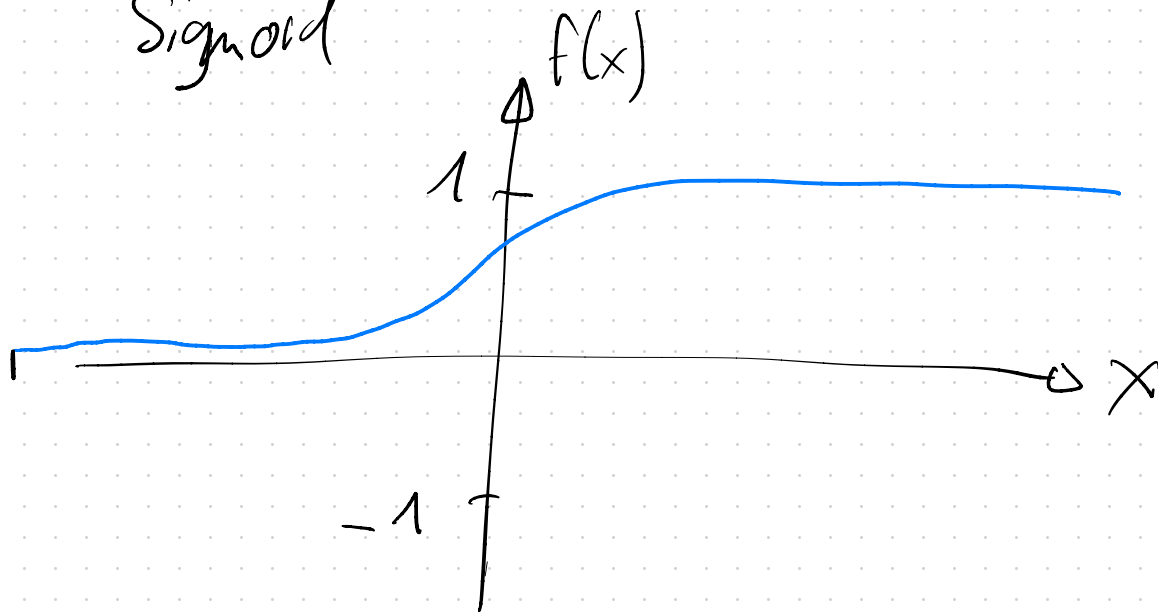
	height (cm)
1	175
2	192
3	152
4	167
5	181

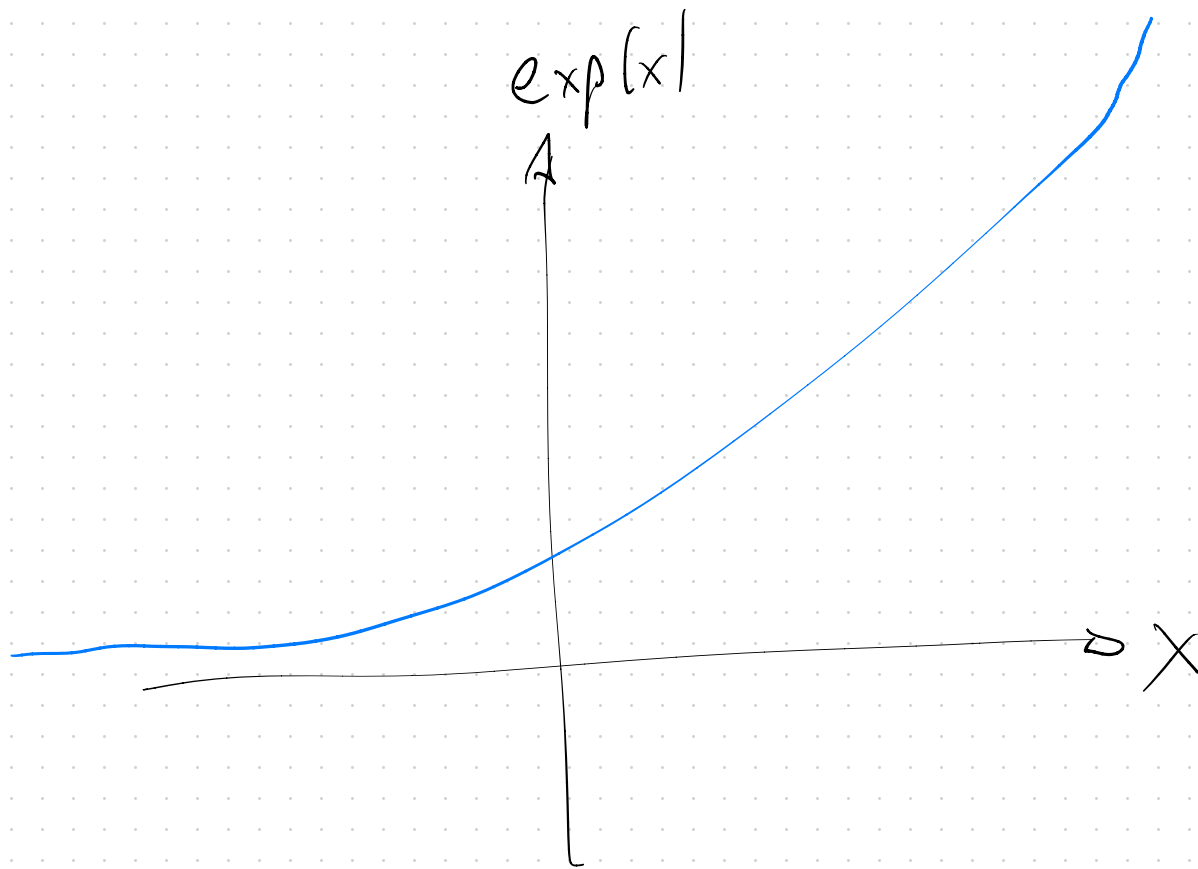


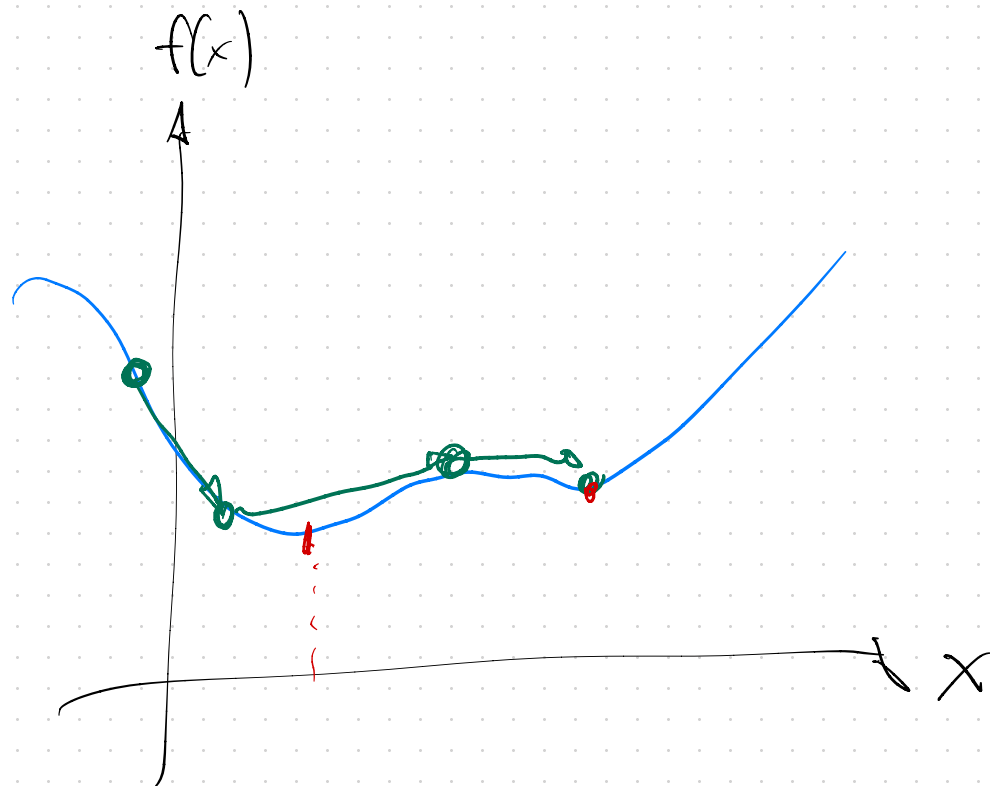
$$p_{\text{model}} = \mathcal{N}(\mu, \sigma)$$

$$\min_{\mu, \sigma} D_{\text{KL}}(p_{\text{model}} \parallel \hat{p}_{\text{data}})$$

Sigmoid







$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}$$

softmax

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix}$$

$$J = \begin{bmatrix} \frac{y_1}{x_1} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}$$

6x6

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\xrightarrow{f(x)} \boxed{y}$$

$$\nabla_x f(x) =$$

$$\begin{bmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \frac{\partial}{\partial x_3} \\ \frac{\partial}{\partial x_4} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x_1} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x_3} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x_4} \end{bmatrix}$$